

Kwame Asante

Research Scientist, Machine Learning

Boston, MA 02115 | (617) 394-2847 | k.asante@email.com

CORE SKILLS

ML/AI: Deep learning, transformer architectures, model evaluation, benchmark design, Python, PyTorch, TensorFlow

Domains: Financial ML, anomaly detection, time-series forecasting, regulatory compliance modeling

Research: Literature review, experimental design, statistical analysis, peer review

Tools & Infrastructure: Jupyter, Weights & Biases, HuggingFace Hub, GPU compute, academic computing clusters

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher

Meridian Financial Research Institute, Boston, MA | Jun 2024 – Present

- Designed and implemented novel deep learning architectures for detecting payment fraud patterns in synthetic transaction datasets; achieved 94.2% AUC on internal benchmark against 89% baseline
- Published research on temporal graph neural networks for transaction risk modeling in *ACM TIST* (accepted 2025); currently at 12 citations in preprint phase
- Evaluated state-of-the-art LLM-based approaches for generating compliance narratives in fintech scenarios; conducted ablation studies on prompt engineering strategies
- Contributed code and benchmark datasets to open-source financial ML repository; maintained documentation for reproducibility

PhD Candidate, Machine Learning

Charles University, Prague | Aug 2020 – May 2024

- Dissertation: "Adaptive Anomaly Detection in High-Frequency Trading Networks" defended May 2024
- Published 4 peer-reviewed papers in top-tier venues:
- "Graph Neural Networks for Market Microstructure" *NeurIPS 2023* (78 citations)
- "Robust Sequence Modeling Under Distribution Shift" *ICLR 2023* (64 citations)
- "Evaluating Transformer-Based Architectures on Payment Flow Data" *ICML Workshop 2022* (31 citations)
- "A Benchmark Suite for Fintech Time-Series Forecasting" *ML for Finance Workshop 2023* (18 citations)
- Designed evaluation framework for anomaly detection models across 7 public financial datasets (Yahoo Finance, Kaggle); published detailed benchmark results comparing 12 baseline architectures
- Developed custom synthetic dataset generator for payment network simulations to enable reproducible anomaly detection research; released on Hugging Face Model Hub

Research Assistant

Nexus AI Lab, Atlanta, GA | Jun 2019 – Jul 2020

- Supported PhD research on time-series classification for financial applications
- Implemented baseline models and conducted literature surveys on deep learning for sequential data
- Assisted with experimental setup and result aggregation across GPU cluster

EDUCATION

PhD in Machine Learning, Charles University, Prague | 2024

Bachelor of Science in Computer Science, University of Ghana, Accra | 2019

SELECTED BENCHMARKS & RESULTS

- *Payment Fraud Detection Benchmark*: Evaluated 8 neural network architectures on curated dataset (150K transactions); top model achieved 96.1% F1 on test set, published methodology in technical report
- *Compliance Text Generation Task*: Tested fine-tuned transformer models (6B parameters) on synthetic regulatory narrative generation; BLEU score 42.3 against human baselines
- Maintained leaderboard for community benchmarking on time-series forecasting; current benchmark aggregates results from 23 international research groups